EX NO: 5	DIODE CLIPPING AND CLAMPING CIRCUITS.	30/7/19

AIM:

A) To Construct positive, negative biased and combinational clippers using diodes.

B) To Observe the waveforms of the positive and negative clamping circuits.

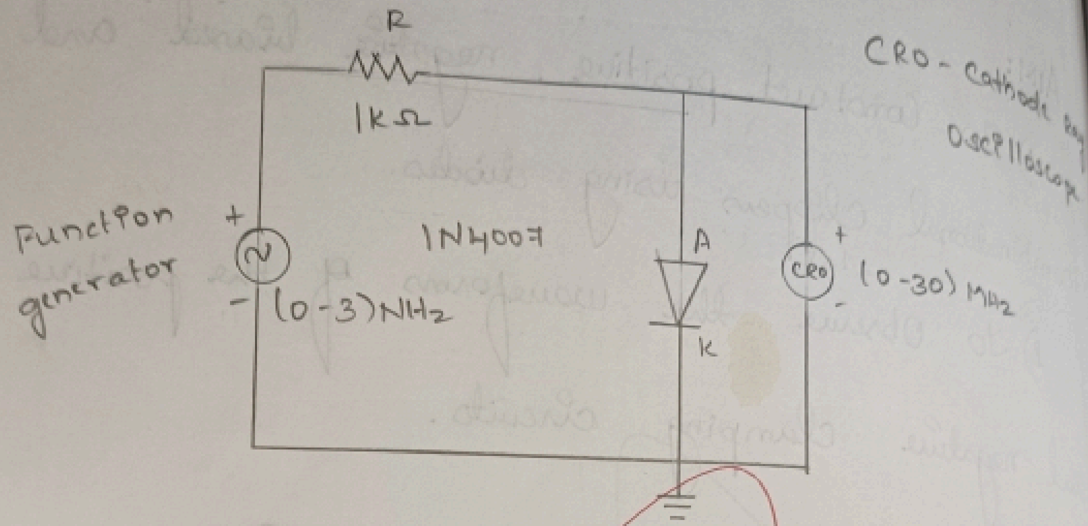
APPARATUS, COMPONENTS AND DEVICES REQUIRED:

S.NO	APPARATUS AND COMPONENTS DEVICES	TYPE / RANGE	QUANTITY
1)	PN Junction diode	IN4007	2
2)	Resistor	1K Ω	2
3)	CRO	(0-30) MHZ	1
4)	Function generator	(0-30) MHZ	1
5)	Dual regulated power supply	(0-30) V	1
6)	Bread Board	–	1
7)	Connecting wires	–	As required

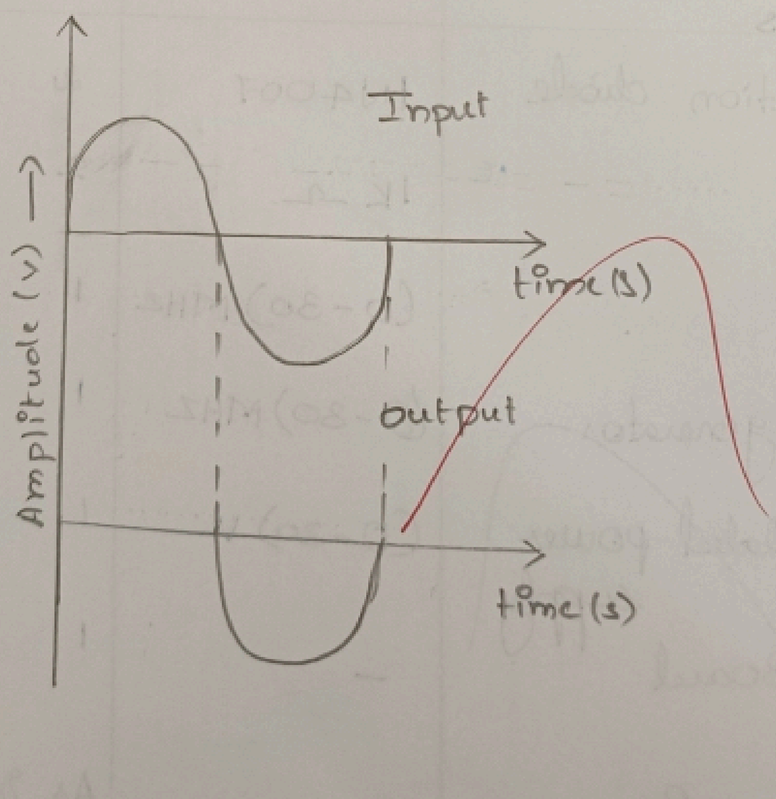
Clippers

Positive clipper:

circuit diagram:



Model Graph:-



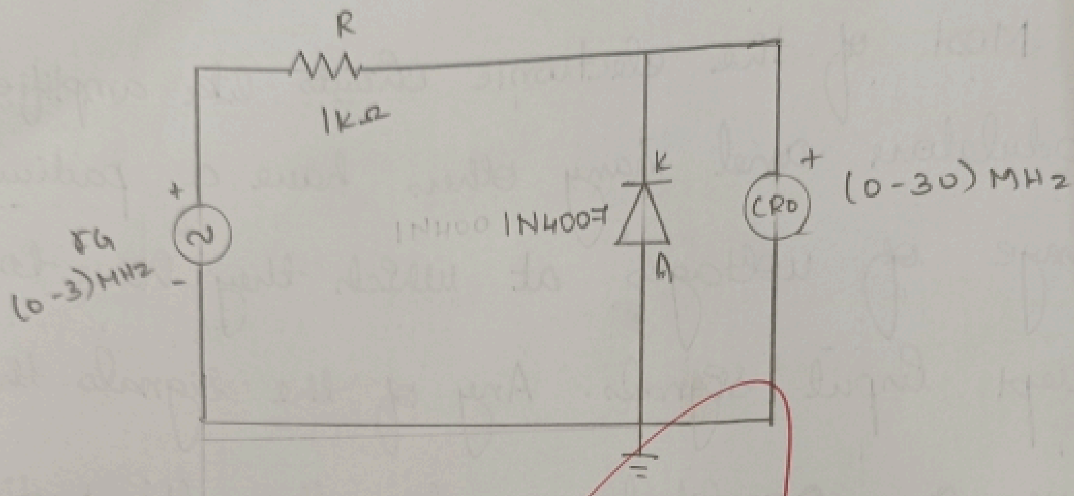
THEORY:

Most of the electronic circuits like amplifier, modulators and many others have a particular range of voltages at which they have to accept input signals. Any of the signals that have an amplitude greater than this particular range may cause distortions in the output of the electronics circuits and may even lead to damage of the circuits components.

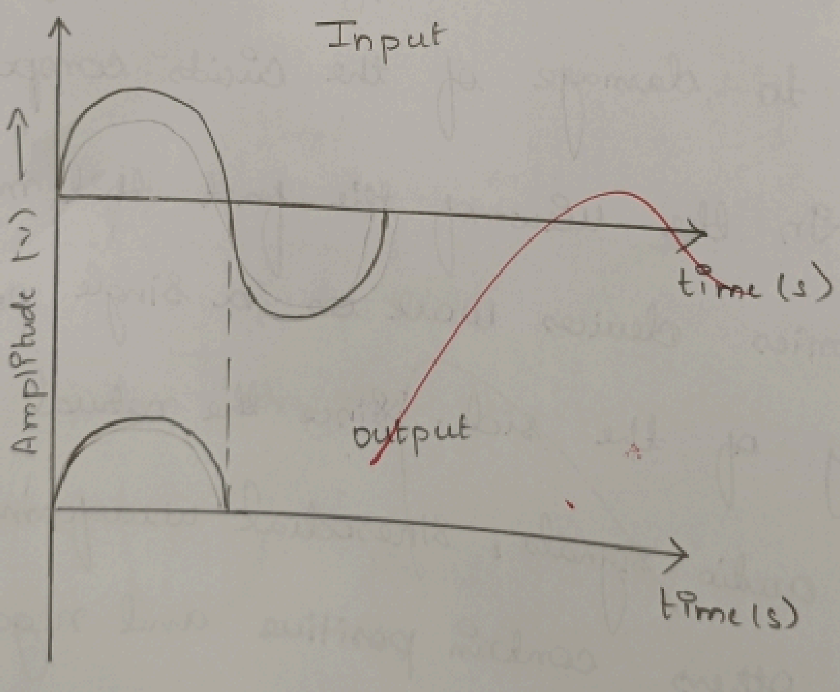
In the view of the fact that most of the electronics devices work on a single positive supply of the side. Since the natural signals like audio signals, sinesodial waveforms and many others contain positive and negative cycles with varying amplitude in their duration.

Negative clipper:

Circuit Diagram:



Model Graph:-



The waveforms and other signals have to be modified in such a way that the signal supply electronic circuits can be able to operate on. The clipping of a waveform is the most common technique that applies the input signals to adopt than so that they may lie with in the clipping of waveforms can be done by eliminating the portions of the waveforms which crosses the input voltage of the circuit.

Clippers can be classified into two basic types of circuits they are: Series clippers and shunt or parallel clippers series clippers contains a power diode is series with the load connected in the end of the circuit the shunt clippers contains a diode in parallel with resistive load.

TABULAR COLOUMN:

POSITIVE CLIPPER				NEGATIVE CLIPPER			
INPUT		OUTPUT		INPUT		OUTPUT	
AMPLITUDE (V)	TIME (s)	AMPLITUDE (V)	TIME (s)	AMPLITUDE (V)	TIME (s)	AMPLITUDE (V)	TIME (s)
2.5V	2x0.5ms = 1ms	-1.2x5V = -6V	2x0.5ms = 1ms	2x5V = 10V	2x0.5ms = 1ms	1.2x5V = 6V	2x0.5ms = 1ms.

The output voltage will be calculated as

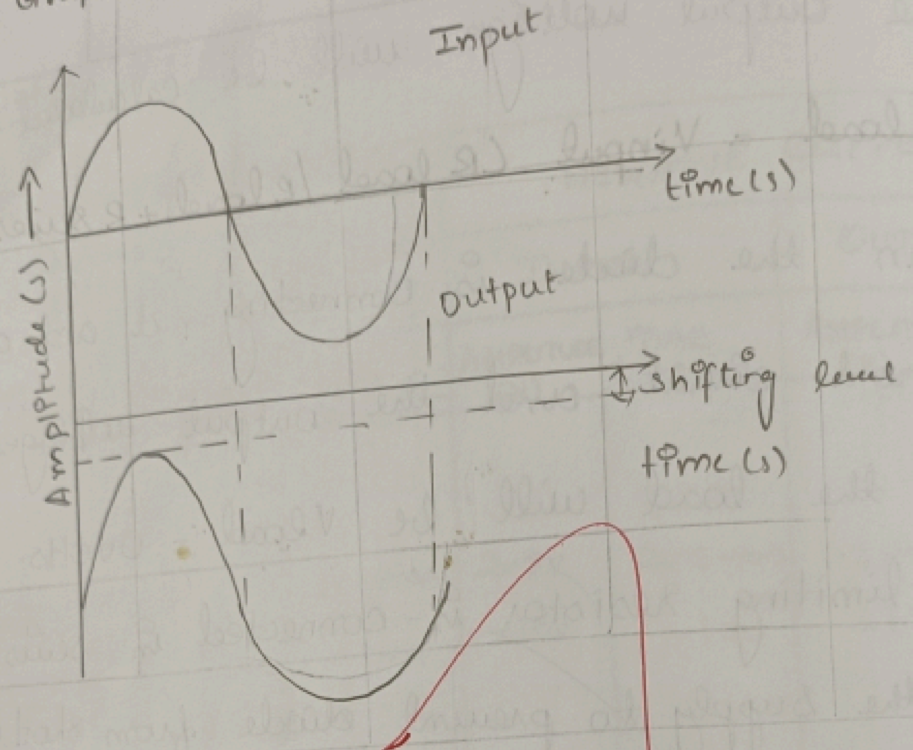
$$V_{load} = V_{input} \, (R_{load} / R_{load} + R_{series}).$$

When the diode is connecting, it acts as a short circuit and the output voltage across the load will be $V_{load} = 0$ volts. The series limiting resistor is connected in series with the supply to prevent diode from short circuits.

In this case, the output voltage of the circuit should be ± 0.7 volts. It depends on the polarity of the shunt clipper which is determined by the direction of diode connection.

The biased clipper circuit is a shunt clipper which uses the DC supply voltage to bias the diode. It is the biasing voltage at which the diode starts conducting. The diode in the shunt clipper circuit starts to conduct.

Model Graph:-



Tabular column:-

CLAMPER	INPUT		OUTPUT	
	AMPLITUDE(V)	TIME (s)	AMPLITUDE(V)	TIME (s)
POSITIVE CLAMPER	2x5V = 10V	0.5 x 2ms = 1ms	2x5V = 10V Shifting level = 1 x 5V = 5V	0.5 x 2ms = 1ms
NEGATIVE CLAMPER	2x5V = 10V	0.5 x 2ms = 1ms	2x5V = 10V. Shifting level = -1 x 5V = -5V	0.5 x -2ms = -1ms

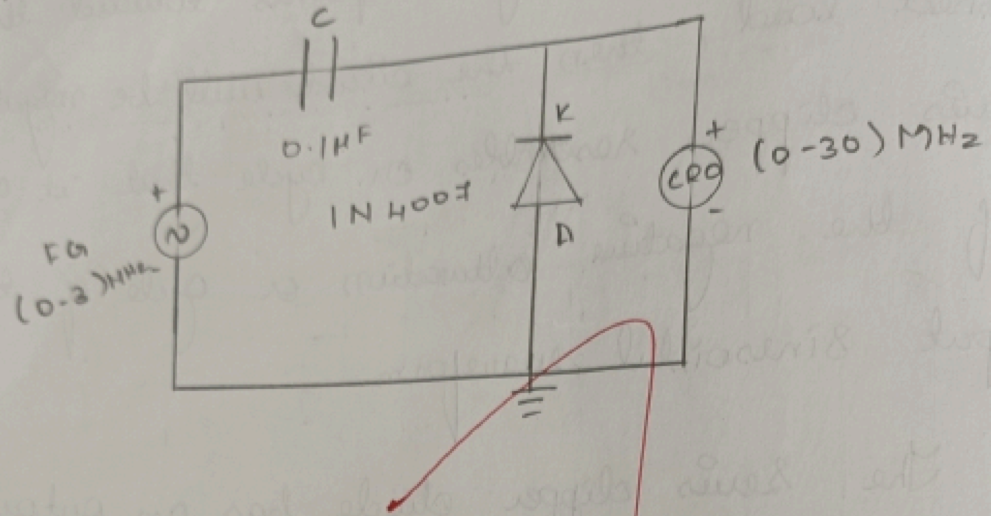
If the diode symbol points toward the connect load, then the circuit will be negative series clipper. resembles or cycle that it clips off the negative alternation or cycle of the input sinusoidal waveform.

The series clipper diode has an output voltage of V load = Vinput when the diode is conducting the input voltage applied by the supply will be dropped and has an output voltage V load = 0 volts.

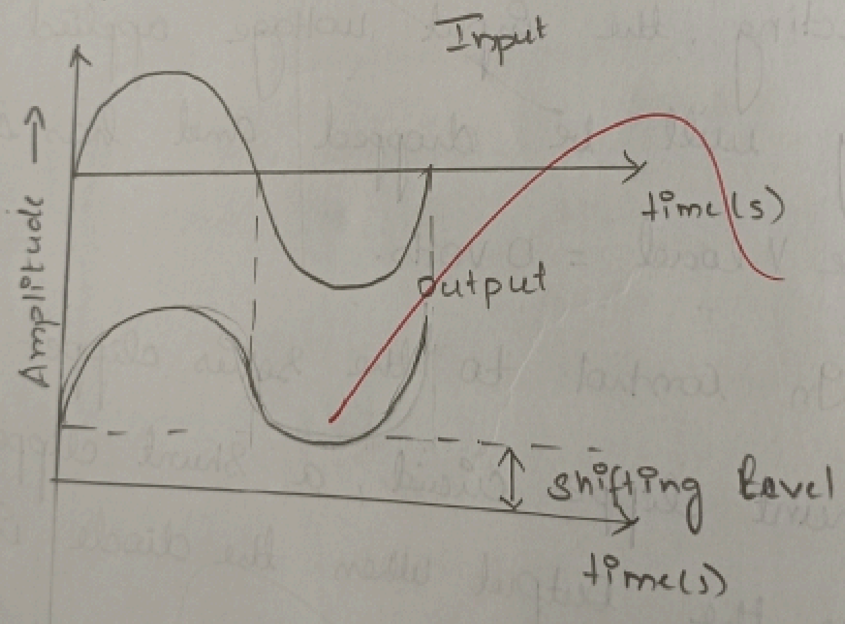
In control to the series clipper circuit, a shunt clipper circuit, a shunt clipper circuit provides the output when the diode is connected in reverse bias and when it is not conducting when the diode is non- conducting. the shunt conducting of the clipper.

Clamper :-

Positive clamper :-



Model Graph :-



1. Altering the waveform shapes.
2. Protecting the circuits from transistors.

A transient is defined as an abrupt change in current or voltage with extremely short duration clipper circuits can be used to the transistor.

PROCEDURE:

Connections are made as per the circuit diagram.

The Input waveform is provided from the function generator.

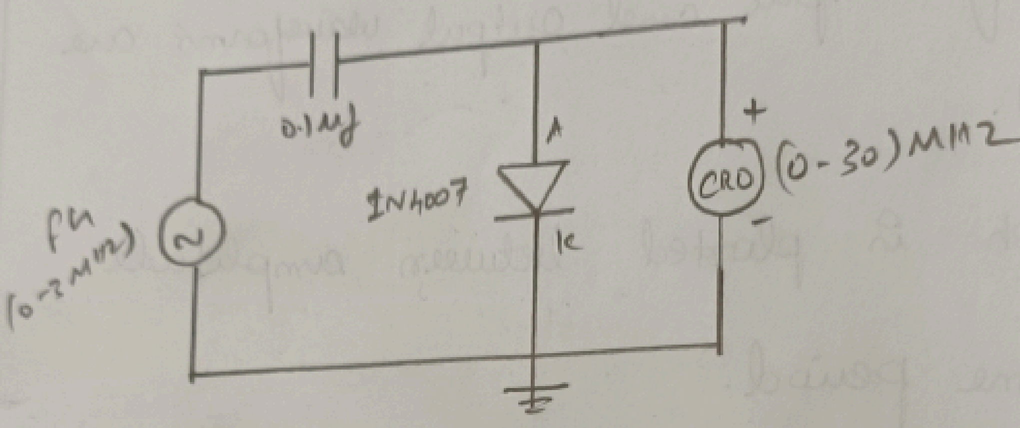
The Output waveform is observed in the CRO.

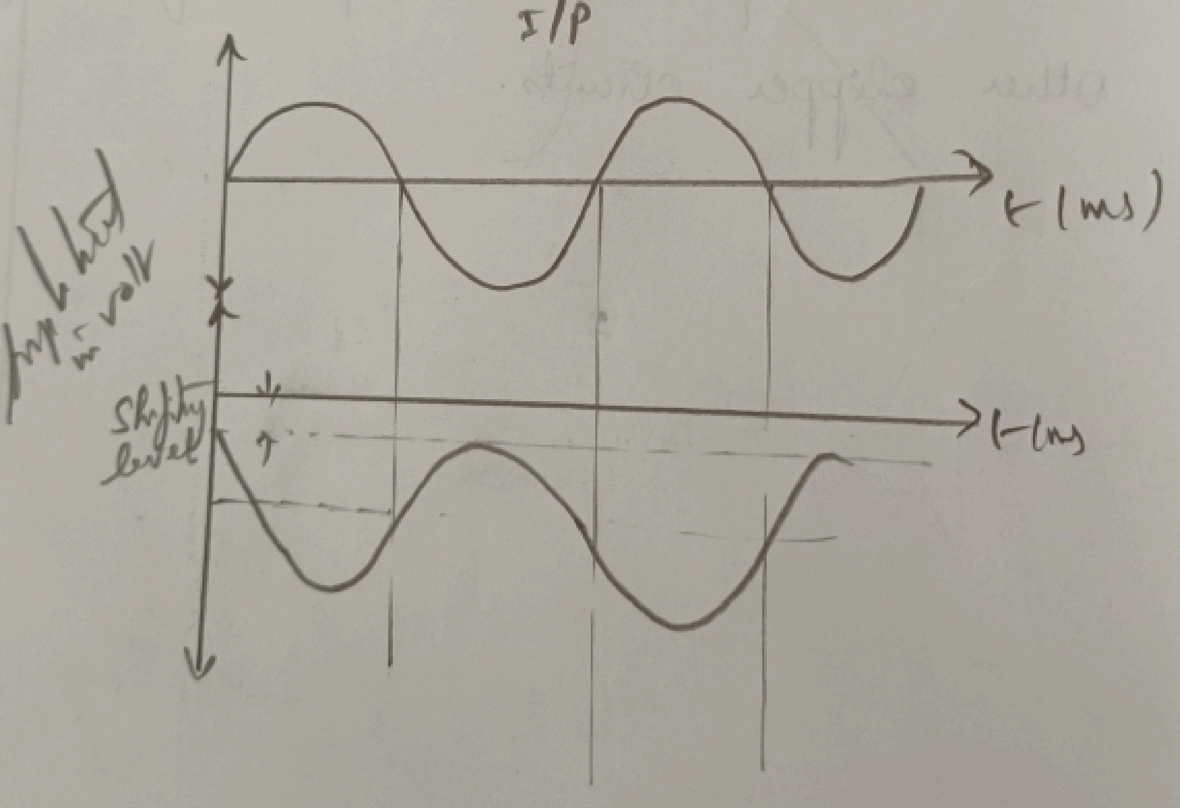
The reading of amplitude and time period for input and output waveforms are tabulated.

Graph is plotted between amplitude and time period.

The above procedure is repeated for the other clipper circuits.

Negative clamper :





No: ....

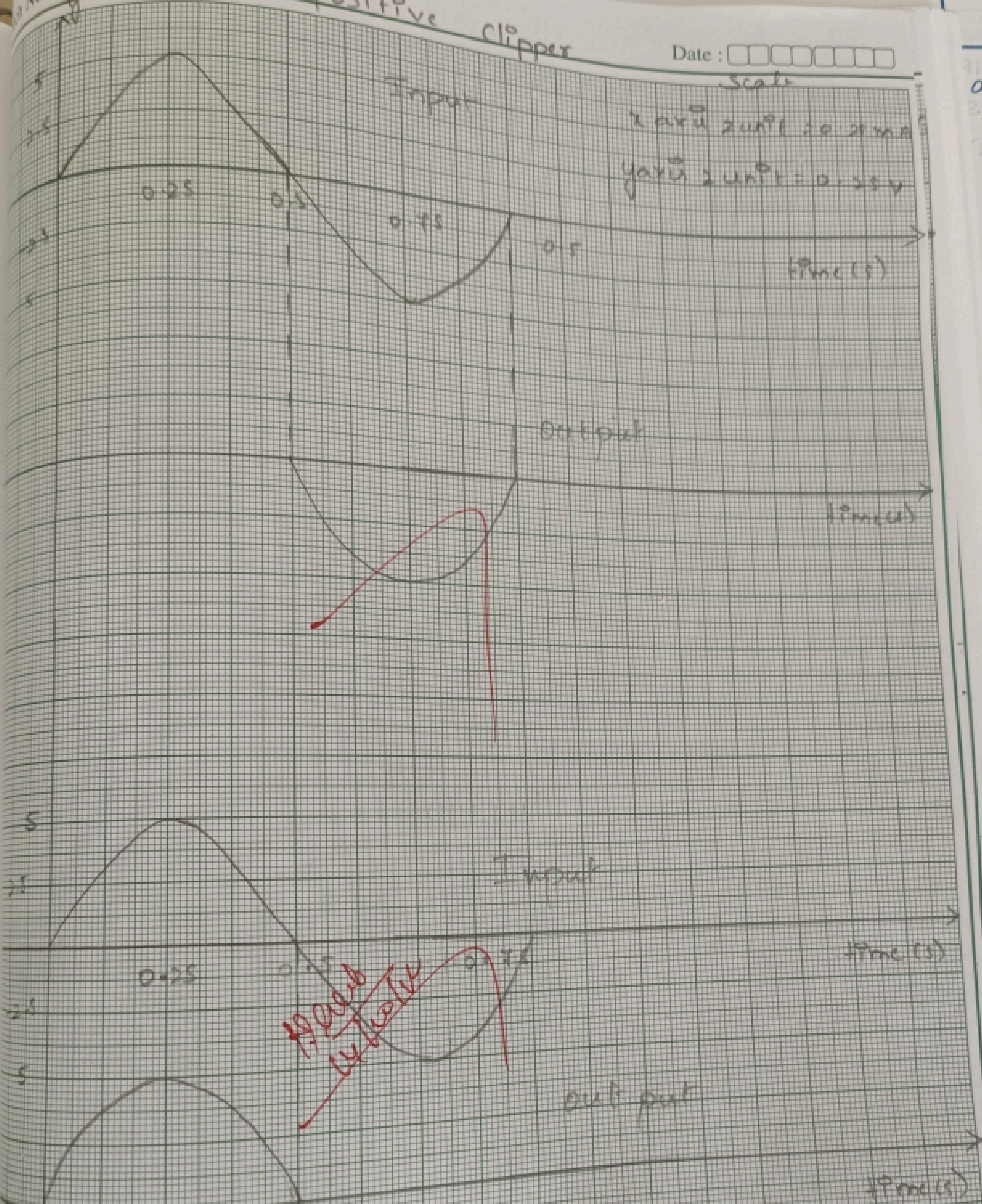
Positive clipper

Date : ☐☐☐☐☐☐☐☐

Scale

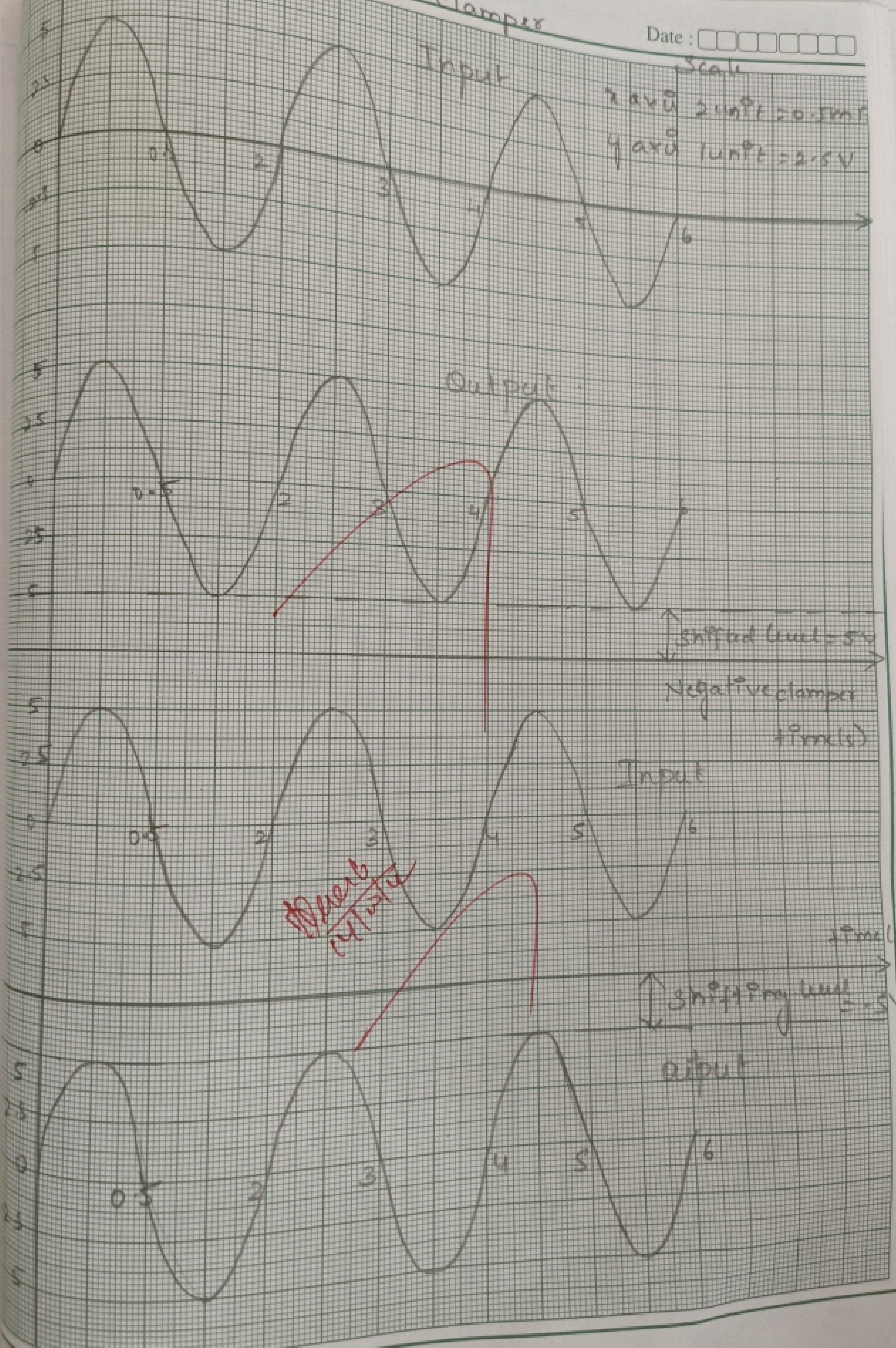
x axis 1 unit = 0.25 ms

y axis 1 unit = 0.25 V



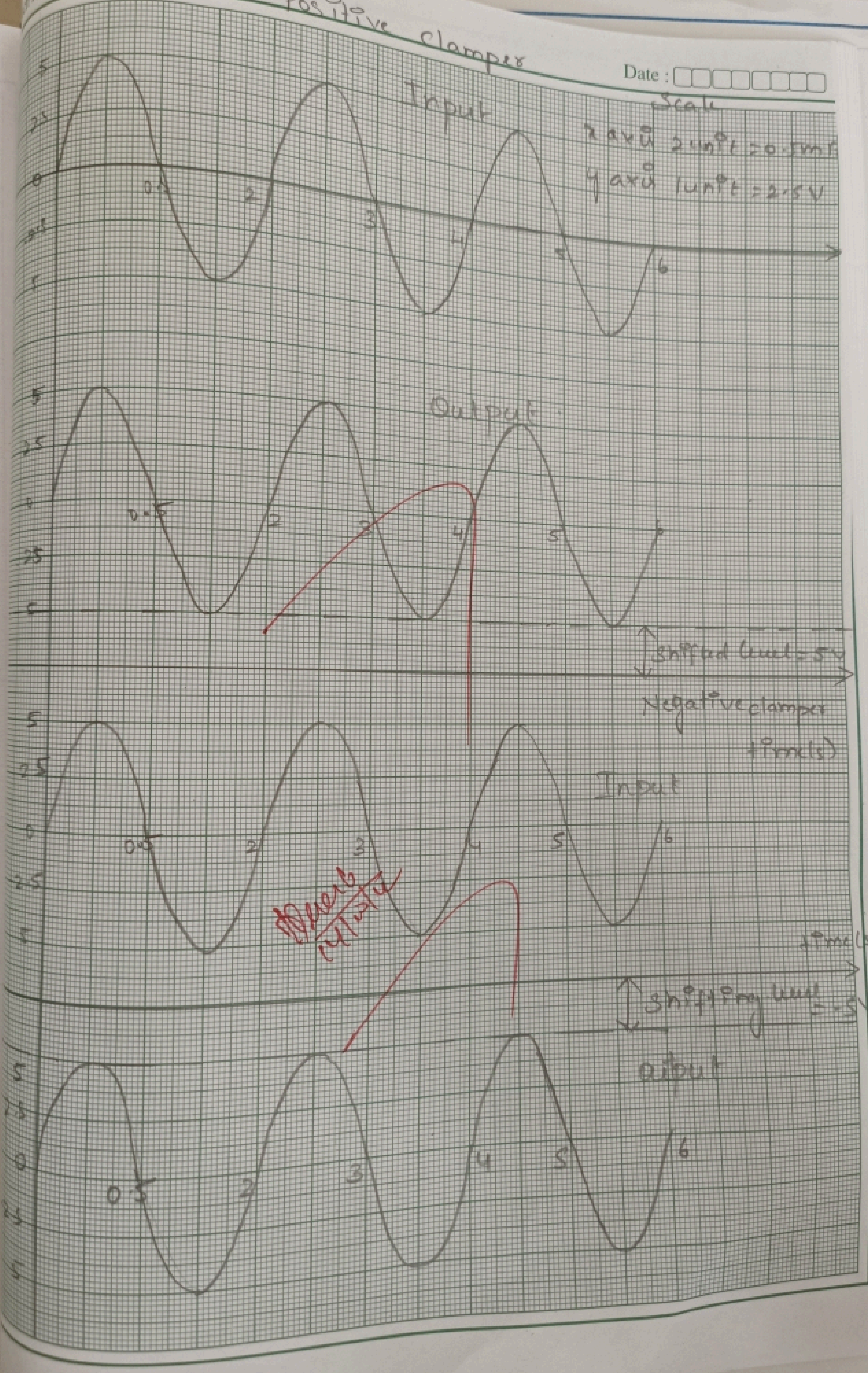
Positive clamper

Date :



Positive clamper

Date :



RESULTS:

Thus the negative, biased and combinational clippers are constructed using diodes and the output waveforms are obtained.

Ex no: 5b	SWITCHING CIRCUIT USING DIODE AND TRANSISTOR.	01.08.17

Aim:

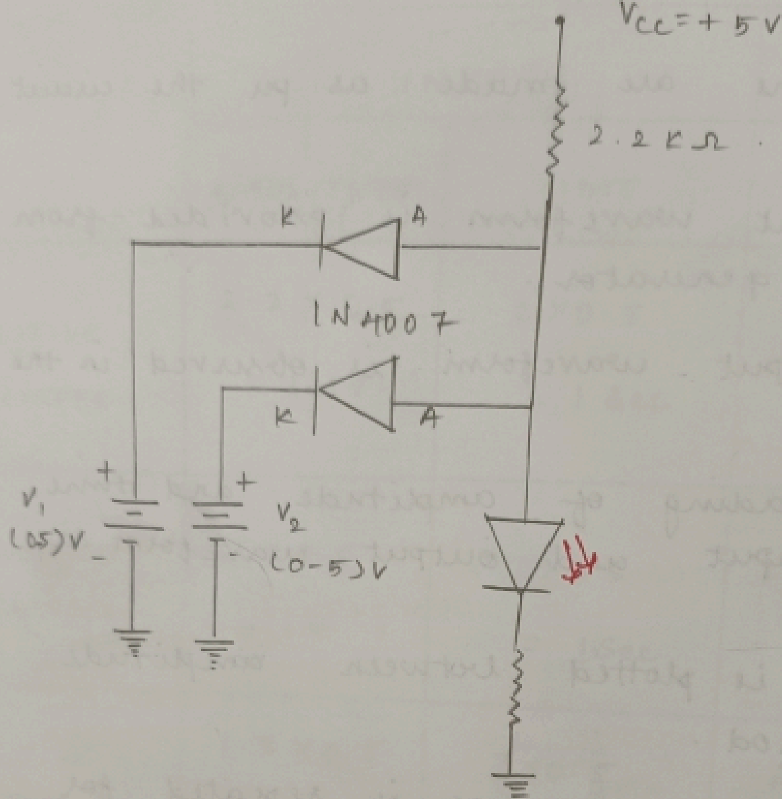
To construct switching circuit using diode and transistor.

* AND and OR logic gates using diodes.
* NOT gate using transistor.

Apparatus Required:

SL NO.	Apparatus / component Name.	Type / Range	Quantity.
1.	Regulated power supply (RPS)	(0-280)V	1
2.	Bread Board	-	1
3.	PN Junction diode	IN4007	2.
4.	Bipolar Junction Transistor (BJT)	BC107	1
5.	Resistor R_1	10KΩ	1
	R_2	15KΩ	1
	R_3	2KΩ	1
6.	connecting wires	-	AS required.

Two input diode AND Gate



Truth Table:

A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

Theory:

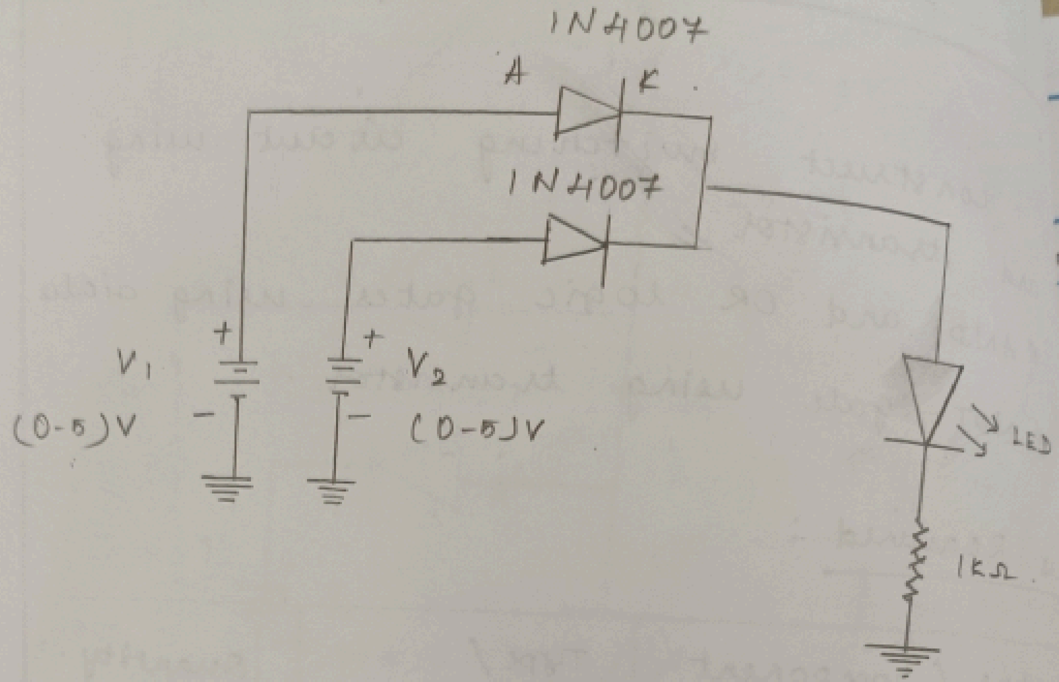
Diode logic gate version

In logic gates logical function are performed by parallel or series connected switches controlled by input logical variables. In diode logic, electrically operated switches are implemented by diodes when forward biased a diode switchs as closed, when backward biased, the switch is open. There are two kind of diode logic gate OR and AND.

OR logic:

OR logic gates are implemented by parallel connected normally open switches. So, in diode OR logic gates, the input voltage source are connected to diode anode. Diode cathod are joined to the output which is connected through the pull-down resistor R_1 to ground. If two diodes OR logic gates are cascaded, they behave as current-sourcing logic gates if the first gate produces high output voltage the second gate consumes current from the first one. If the first gate produces

Two input diode OR Gate



Truth table

A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

low output voltage, the second gets gate does not injected current into the output of the first one. A diode OR gate does not use its own power supply. The input source with high voltage (logical 1) supply the load through the forward biased diodes.

AND logic:

The diodes are reverse connected and forward biased by an additional voltage source +V through the pull-up resistor R_1. The input voltage source are connected in opposite direction to the supplying voltage source (travelling along the loop +V - R1 - D - Vin). To invert the output voltage and to get a grounded output. The complementary voltage drop (+V - V_{R1}) between the output and ground the input source. The diode is forward biased (the diode switch is closed) and the output voltage drop accross the diode is low (output logicall 0) The output resistance is low and is determined by the input source. The rest of diodes connected to high input voltages (input logical 1s) are backward biased and their input source are disconnected from the output.

If two diode AND gates are connected cascaded, they behave as current. sinkining logic gates if the first gate produces high output voltage, the second gate does not consume current from the first once if the first gate produces low output voltage, the second gate injects current into the output of the first one. A diode AND gate uses its own power supply to drive the load through the pull-up resistor is taken as an output instead the floating voltage drop V_{R1} across the resistor.

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RESULT:

Thus the following switch circuit are constructed using diode and transistor

a) AND and OR logic gates using diodes.